A Systematic Comparison of Multimodal Fusion Strategies for Cross-Region Building Damage Assessment Using Optical and SAR Imagery

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Abstract

Multimodal remote sensing is an important direction for post-disaster building damage assessment because optical and SAR imagery provide complementary information, but it remains unclear which fusion strategy is most suitable for robust cross-region generalisation. In this project, we systematically compare early, mid-level, and late fusion within a shared Transformer-based architecture family. The dual-branch mid-level and late-fusion models use architecturally identical but independently parameterised optical and SAR encoders. We also evaluate the same architecture family under a stronger training recipe with modality-specific normalisation, synchronised augmentation, Lovasz-based losses, and joint localisation–damage supervision. Experiments on the BRIGHT earthquake subset use a cross-region split with training on Turkey and Morocco, validation on Noto, and testing on the unseen Haiti earthquake. The results show that fusion strategy matters: Late Fusion performs best overall on Haiti, while Early Fusion is consistently the weakest. Mid-level fusion remains a meaningful design to study, but its additional modules do not show stable gains across settings, and the stronger training recipe changes absolute performance without consistently favouring the more complex model. Per-class evaluation further shows that all variants struggle on the minority damage categories under cross-region shift. Overall, the main value of the project lies in the systematic comparison across fusion settings and ablations, showing that extra architectural complexity does not automatically lead to more robust damage assessment.

1. Introduction

Accurate and timely information about building damage is essential in the first stages of post-disaster response. After a major earthquake, emergency teams need to know not only where buildings are located, but also which areas are likely to contain severe structural damage, so that rescue efforts and resource allocation can be prioritised effectively. Satellite imagery is particularly useful in this context because it can provide rapid, large-scale coverage when ground surveys are slow or difficult to organise. Among the available modalities, optical and synthetic aperture radar (SAR) imagery offer different but complementary views of the same scene. Optical images capture visible structural details, while SAR remains informative even under cloud cover, smoke, or poor lighting conditions. This makes multimodal damage assessment a practical and important problem for disaster mapping (Chen et al., 2025; 2022; Li et al., 2022).

Despite this potential, combining optical and SAR information effectively is still not straightforward. In multimodal remote sensing, performance often depends not only on the backbone architecture but also on where and how the modalities are fused (Li et al., 2022). If the two signals are merged too early, modality-specific cues may be weakened before meaningful representations have been learned. If they are kept separate until the end, potentially useful cross-modal interactions may be underused. A further challenge is that damage patterns are rarely regular. Collapsed roofs, partial structural failures, and debris fields often appear fragmented and geometrically inconsistent, which motivates interest in more adaptive feature extraction modules such as deformable convolution (Zhu et al., 2019). These issues motivate our study: rather than assuming one fusion strategy is best in advance, we ask how early, mid-level, and late fusion compare under a shared architecture and whether extra adaptive components within mid-level fusion provide robust gains.

Our work is inspired by the dual-task Transformer architecture introduced for building damage assessment (Chen et al., 2022), but adapts its design for a controlled multimodal comparison. Our dual-branch variants use independently parameterised optical and SAR encoders. This choice reflects the heterogeneous nature of the two modalities and allows each branch to learn modality-specific representations before fusion. We implement early, mid-level, and late fusion within a common experimental framework, and we further analyse the mid-level design through two targeted ablations: one that removes gated attention and another one that replaces deformable residual blocks with standard residual blocks. This allows us to separate the question of when multimodal information should be fused from the question of how an intermediate fusion pathway should be designed. In addition to the main shared cross-entropy training setup, we also evaluate the same architecture family under an extended recipe with modality-specific normalisation, synchronised augmentation, Lovasz-based losses, and joint localisation–damage supervision. This second set of experiments helps us examine whether apparent architectural advantages remain stable once the optimisation recipe changes.

We place particular emphasis on evaluation under geographic separation. Instead of relying on a conventional random split, we reorganise the BRIGHT dataset into a cross-region setting in which the training, validation, and test sets come from different disaster regions (Chen et al., 2025). Models are trained on imagery from Turkey and Morocco, validated on Noto, Japan, and finally tested on the unseen Haiti earthquake set. This setup is more demanding than a random split, but it better reflects the practical scenario in which a model trained on past events must generalise to a newly affected region with different building layouts, materials, and damage characteristics.

The results are interesting as they break one of our original expectations that "stronger architecture is better". Across both training setups, fusion strategy matters: early fusion is consistently the weakest, while late fusion is the strongest overall on the unseen Haiti test set. Mid-level fusion design remains worthy analysing, but additional components to it do not show stable gains. In the simpler shared cross-entropy setting, the no-deform variant outperforms the full mid-level model, while in the extended recipe the ranking becomes more compressed and the no-gating variant becomes more competitive. This suggests that optimisation choices may be as important as the architectural refinements themselves. Moreover, per-class evaluation shows that all variants struggle on the minority damage categories, with predictions concentrating almost entirely on background and no-damage pixels under cross-region shift. Taken together, the main contribution of this project is to make the systematic comparison across fusion settings and ablations, and reveal that architectural complexity does not naturally translate into more robust damage assessment.

2. Related Work

Building damage assessment from remote sensing imagery has become an important research topic in post-disaster response because it enables rapid large-scale damage mapping when field surveys are slow or difficult to conduct. Early progress in this area was closely tied to the development of benchmark datasets and supervised pipelines based mainly on optical imagery. In particular, xBD established a large-scale benchmark for building localisation and damage classification, and helped standardise the use of deep neural networks for post-disaster building damage assessment (Gupta et al., 2019). Subsequent work moved beyond low-level change detection and increasingly emphasised building-aware or object-aware damage mapping, showing that preserving building structure is important for producing useful damage predictions. While these studies significantly advanced the task formulation, they remain largely centred on optical imagery and therefore do not fully address the complementary role of heterogeneous sensing modalities under adverse imaging conditions.

As model architectures improved, the literature gradually shifted from conventional convolutional networks toward Transformer-based designs that can capture richer spatial context. Swin Transformer introduced a hierarchical vision Transformer with shifted windows and has since become a strong backbone for dense prediction tasks (Liu et al., 2021). In the specific setting of post-disaster building damage assessment, Transformer-based Siamese frameworks have also been proposed to jointly model building localisation and damage classification (Chen et al., 2022). More recent work continues this direction by using hierarchical Transformer encoders to strengthen feature representation and contextual modelling. However, although these approaches improve the encoder side of the problem, they do not by themselves resolve the separate question of how different sensing modalities should be fused within the network.

A related line of work studies multimodal remote sensing fusion more broadly. Existing reviews commonly distinguish among early fusion, intermediate fusion, and late fusion, which differ in when cross-modal interaction occurs and how much modality-specific information is preserved (Li et al., 2022). This distinction is especially relevant for disaster response, where optical and SAR imagery provide complementary information. Optical images contain intuitive appearance and texture cues, while SAR remains informative under cloud cover, smoke, or poor illumination. This complementarity is now reflected in recent multimodal benchmarks such as BRIGHT, which provides a globally distributed dataset for building damage assessment using very-high-resolution optical and SAR imagery (Chen et al., 2025). At the same time, the availability of multimodal data does not by itself determine how fusion should be performed. Controlled comparisons of early, mid-level, and late fusion for building damage assessment remain limited, especially when the backbone family and optimisation setup are kept consistent across variants.

Another issue that has received increasing attention is generalisation beyond the training distribution. Many building damage assessment models are evaluated under relatively homogeneous settings in which training and test samples come from similar events or regions. This can overestimate practical deployment performance. More recent work has started to consider transferability across disasters or domains, motivated by the fact that newly affected regions may differ substantially in building materials, urban layouts, and damage patterns. This direction is important because it reflects the real deployment scenario more closely, but it also leaves open a complementary question: whether different multimodal fusion strategies lead to different levels of robustness under geographic shift.

Our work is positioned at the intersection of multimodal fusion and cross-region generalisation for post-disaster building damage assessment. Relative to optical-centric studies, we explicitly consider the multimodal setting of pre-event optical imagery and post-event SAR imagery. Relative to prior Transformer-based approaches, our focus is not simply to adopt a stronger encoder, but to compare early, mid-level, and late fusion within a unified Swin Transformer-based architecture family with independently parameterised optical and SAR encoder branches in the dual-branch variants. We further analyse the mid-level design through targeted ablations of gated attention and deformable residual blocks, and we evaluate all variants under a geographically separated split across earthquake regions. Therefore, our contribution is to clarify which fusion choices remain beneficial, which added components do not show stable gains, and how these design decisions interact with cross-region robustness.

3. Architecture

Our models are inspired by the dual-task Transformer framework proposed for building damage assessment (Chen et al., 2022), but they are not Siamese networks in the strict parameter-sharing sense. A Siamese network normally uses two encoder branches with the same weights. This is reasonable when the two inputs have the same type, such as pre- and post-event optical images, because the model can compare representations produced in a common feature space. Our task instead combines pre-event optical imagery with post-event SAR imagery. These modalities differ in their appearance, statistical distributions, and useful low-level cues. For this reason, the mid-level and late-fusion variants use two independently parameterised Swin encoders: the encoders have the same architecture, but they do not share weights. This allows each branch to learn modality-specific representations before fusion. Early Fusion uses one Swin encoder after the optical and SAR inputs have been concatenated, so it does not contain two encoder branches.

To support a controlled comparison across early, mid-level, and late fusion settings, we keep the encoder family, decoder structure, output tasks, and training protocol as consistent as possible across variants. The key difference between the variants is therefore when and how optical and SAR information is combined (Li et al., 2022).

Across all variants, we use Swin Transformer feature extractors and aggregate multi-scale features from four stages. The deeper feature maps are upsampled to the spatial resolution of the shallowest stage and concatenated channel-wise, producing a 1440-channel representation for each branch in the dual-branch variants. All models output both a binary building localisation map and a four-class damage map. This common output structure allows the same architecture family to be evaluated both under the main cross-entropy setup and under the extended training recipes considered later.

3.1. Fusion Settings Compared in This Study

To systematically compare multimodal integration strategies, we evaluate three main fusion settings: early fusion, mid-level fusion, and late fusion. For the two-branch settings, "separate encoders" means that the optical and SAR branches are architecturally identical Swin encoders with independent parameters; no weights are shared between modalities.

Early Fusion: Optical and SAR images are concatenated at the input stage. Since the combined input has 6 channels, a 1×1 convolution first projects it to 3 channels to match the expected input shape of the ImageNet-pretrained Swin Transformer, before it is passed through one Swin encoder. The resulting joint representation is then used for both localisation and damage prediction. This design is simple, but it mixes the two modalities before modality-specific patterns have been learned.

Mid-Level Fusion: Optical and SAR inputs are first processed by two separate, independently parameterised Swin encoders. Their aggregated 1440-channel representations are then used differently: the optical features are passed to the localisation head, while the damage head operates on a fused optical–SAR representation. This design preserves modality-specific processing at the encoder stage while still allowing multimodal interaction before the final prediction.

Late Fusion: Optical and SAR inputs are kept separate through both feature extraction and head-level processing by two independently parameterised Swin encoders. In our implementation, the optical branch produces both localisation and damage predictions, while the SAR branch produces a damage prediction. The two damage prediction tensors are concatenated and combined by a final 1×1 convolution to obtain the final damage map. This delays multimodal interaction until the decision stage.

3.2. Mid-Level Fusion Design

Since mid-level fusion remains worth examining , we further examine two components within this setting: a gated attention fusion module and deformable residual blocks.

3.2.1. Gated Attention Fusion

We define a custom *gated channel fusion* module (Hu et al., 2018). Let $\mathbf{F}_{opt}$ and $\mathbf{F}_{sar}$ denote the feature tensors extracted from the optical and SAR branches. Each tensor has 1440 channels at the fusion stage in our implementation. We first concatenate them along the channel dimension:

$$\mathbf{F}_{cat} = \text{Concat}(\mathbf{F}_{opt}, \mathbf{F}_{sar}),$$

this produces a fused tensor with 2880 channels.

Instead of directly passing this concatenated representation to the following layers, we introduce a learnable gating mechanism. This mechanism generates an attention mask over the concatenated channels and uses that mask to recalibrate the fused representation. The concatenated tensor is passed through two 1×1 convolutional layers with a bottleneck structure of:

$$\mathbf{A} = \sigma\left(W_2\,\phi\left(\text{BN}(W_1\mathbf{F}_{cat})\right)\right),$$

where W_1 and W_2 denote the learnable 1×1 convolutions, $\text{BN}(\cdot)$ is batch normalization, $\phi(\cdot)$ is the ReLU activation function, and $\sigma(\cdot)$ is the sigmoid function.

The role of the first convolution is to learn an intermediate representation of the concatenated multimodal features. The second convolution then projects this representation back to the original channel dimension. The sigmoid activation then constrains the gating values to the range $[0, 1]$ and thus, they can be interpreted as soft importance weights. The recalibrated fused representation is obtained by element-wise multiplication:

$$\mathbf{F}_{fused} = \mathbf{F}_{cat} \odot \mathbf{A}.$$

The network is thus encouraged to learn how much of the concatenated optical–SAR representation should be preserved at each channel, rather than treating all channels as equally informative. Since the gating operation is implemented with 1×1 convolutions it does not alter the spatial resolution of the feature maps.

Later in the paper, we test this design choice through ablation.

3.2.2. Deformable Residual Fusion Blocks

In addition to the gated channel fusion module, we modify the residual fusion blocks used after feature aggregation. Our motivation is that building damage patterns are often irregular. Damaged structures do not always follow clean patterns. They may appear as partially collapsed roofs or scattered debris. A standard convolution detects features on a regular grid and this may limit its ability to adapt to such irregular patterns.

To address this limitation, we replace the second standard convolution in the original residual block with a deformable convolution (Zhu et al., 2019). We put deformable convolutions in the task-specific heads. They are applied to the unimodal optical features for building localization, and to the fused multimodal features for damage classification. This allows the network to adapt to irregular building shapes and scattered debris patterns respectively. The first convolution in the block is kept unchanged. This part of the block still performs a standard local transformation, followed by batch normalization and ReLU activation. An additional convolutional layer predicts spatial offsets after this initial transformation. These offsets are then used by the deformable convolution to adapt the sampling positions of the convolution kernel. Instead of collecting information from a fixed 3×3 neighbourhood, the block can focus on more informative locations. Let $\mathbf{X}$ denote the input feature map. A standard convolution computes responses over a fixed sampling grid. Deformable convolution on the other hand augments each sampling location with a learned offset Δp_n so that the output at position p_0 is computed as

$$\mathbf{Y}(p_0) = \sum_{n=1}^{N} w_n\,\mathbf{X}(p_0 + p_n + \Delta p_n),$$

where p_n denotes the regular sampling locations of the convolution kernel, Δp_n are the learned offsets, and w_n are the convolution weights (Zhu et al., 2019).

As with gating, the contribution of this design choice is evaluated later through ablation.

3.3. Mid-Level Ablation Variants

To understand which parts of the mid-level design matter most, we also evaluate two ablated variants.

Mid-Level (No Gating): This variant replaces the gated fusion module with a plain concatenation-based projection block. The concatenated optical–SAR tensor still passes through a bottlenecked 1×1 projection so that tensor shape and intermediate dimensionality remain comparable to the gated version, but without sigmoid gating or element-wise reweighting.

Mid-Level (No Deform): This variant keeps the gated fusion module unchanged but replaces deformable residual blocks with standard residual blocks in the task-specific heads.

Together, these variants let us separate the effect of how optical and SAR features are fused from the effect of how the fused features are refined afterwards. The same five architecture variants are used both in the main cross-entropy comparison and in the extended recipe experiments, which helps keep the later analysis consistent.

4. Training Methods

To keep the comparison fair, all five architecture variants were trained under shared optimisation settings. This includes the three main fusion settings (early, mid-level, and late fusion) and the two mid-level ablation variants. Our main goal is to compare architectural choices under a common training protocol, and then examine whether stronger training recipes alter the same ranking.

All experiments were run on an NVIDIA L4 GPU. Because the models are relatively memory-intensive, especially in the dual-branch settings, we used a batch size of 2 throughout. We also used Automatic Mixed Precision (AMP), which reduces memory usage and speeds up training while preserving numerical stability (Micikevicius et al., 2018). A fixed random seed of 456 was used for all runs.

The data split was fixed by geographical region. Training was carried out on samples from the Turkey and Morocco earthquakes, validation on Noto, and final testing on Haiti. In the main setup, optical pre-event imagery was converted to floating-point tensors and scaled to the range $[0, 1]$. Since the post-event SAR inputs are single-channel, they were repeated across three channels so that they could be processed by the same Swin-based feature extraction pipeline.

4.1. Main Comparison Protocol

For the learning process, we used the AdamW optimizer and set the learning rate to 0.0001 along with a weight decay of 0.0001. We specifically chose AdamW because foundational research shows it is significantly better at training Transformer models like our swin transformer compared to older optimizers (Loshchilov & Hutter, 2019; Liu et al., 2021)

We used cross-entropy as the loss function. Since our main goal is to classify each building into specific damage levels, cross-entropy is the standard and most proven mathematical tool for this type of multi-class categorization in satellite imagery (Gupta et al., 2019):

$$\mathcal{L}_{main} = \mathcal{L}_{CE}(\hat{Y}_{dam}, Y_{dam}).$$

Although each model also outputs a localisation map, the current baseline implementation applies supervision only to damage classification. This keeps the main comparison focused on fusion design and it avoids introducing additional loss-balancing choices into the architectural study.

After every epoch, validation loss was computed on the held-out validation set. If the validation loss improved, the model weights were saved. At last, We report results from the best validation checkpoint rather than from the final epoch.

4.2. Extended Training Recipe Experiments

To examine whether the conclusions are sensitive to the optimisation setup, we also ran an extended set of experiments using the same five architecture variants but with a stronger training recipe. These experiments are supplementary to the main comparison.

In this extended setup, optical inputs were normalised using ImageNet mean and standard deviation to better match the Swin-T pretraining distribution, while SAR inputs were normalised independently by min–max scaling. Training data also used synchronised spatial augmentation across optical images, SAR images, and target masks, including random horizontal and vertical flips, 90° rotations, and random 512×512 crops. Validation and test data were not augmented.

The extended experiments also used joint supervision for localisation and damage prediction, with a combined cross-entropy and Lovasz-Softmax objective:

$$\mathcal{L}_{ext} = \mathcal{L}_{CE}^{dam} + \mathcal{L}_{CE}^{loc} + 0.75\,\mathcal{L}_{Lovasz}^{dam} + 0.5\,\mathcal{L}_{Lovasz}^{loc}.$$

The localisation target is derived from the damage mask by treating all non-background pixels as building pixels. The Lovasz terms were included because they directly optimise overlap-based segmentation quality, which is closely related to IoU.

These extended experiments were again trained for 20 epochs with AdamW (10^{-4} learning rate, 10^{-4} weight decay), but now with a cosine annealing learning-rate scheduler. As in the main comparison, the best model checkpoint was selected according to validation loss. By keeping the architecture family fixed while changing the training recipe, we can more clearly separate effects caused by fusion design from those caused by optimisation choices.

5. Numerical Results

5.1. Evaluation Setup and Metrics

Our numerical evaluation focuses on cross-region generalisation. Rather than using a random train–test split, we evaluate how well each model transfers to a geographically unseen disaster region. As described in Section 4, training is carried out on Turkey and Morocco, validation on Noto, and final testing on Haiti. This setting is more demanding, but it better reflects the practical scenario in which a model trained on previous disasters must be applied to a newly affected region.

We evaluate five architecture variants: Early Fusion, Mid-Level (Full), Mid-Level (No Gating), Mid-Level (No Deform), and Late Fusion. The main quantitative metrics are Macro F1, mean Intersection over Union (mIoU), Pixel Accuracy, and Cohen's κ. Macro F1 is especially important here because it reduces the risk of over-interpreting results that are dominated by the majority classes. We also use per-class IoU and F1, qualitative visual examples, and training curves to better understand where the models succeed and where they fail. Enlarged versions of the figures shown in this section, together with per-class numerical breakdowns and the extended-recipe training curves, are provided in the Appendix.

5.2. Main Comparison Under the Shared Cross-Entropy Protocol

We first compare all five variants under the shared cross-entropy protocol introduced in Section 4. The results on the Haiti test set are shown in Table 1.

Table 1. Main comparison under the shared cross-entropy protocol on the unseen Haiti test set.

Model	Macro F1	mIoU	Pixel Acc.	Kappa
Late Fusion	**0.4349**	**0.3914**	**0.9403**	**0.7396**
Mid-Level (No Deform)	0.4235	0.3759	0.9301	0.6941
Mid-Level (Full)	0.4072	0.3564	0.9277	0.6296
Mid-Level (No Gating)	0.3895	0.3363	0.9178	0.5602
Early Fusion	0.3339	0.2626	0.7498	0.3847

From table 1, we can find that fusion strategy matters. Late Fusion gives the strongest overall performance on the test set, while Early Fusion is consistently the weakest. This suggests that, in the current cross-region setting, keeping optical and SAR processing separate until the decision stage is more robust than mixing them too early. Early Fusion is not only the lowest-scoring model in Macro F1 and mIoU, but also falls far behind in Pixel Accuracy and Cohen's κ, indicating that its errors are not limited to a small number of difficult pixels.

Besides, the mid-level design provides a useful setting for analysis, but the full version is not the best-performing model. Mid-Level (No Deform) outperforms the full mid-level model across all four metrics, while Mid-Level (No Gating) performs worse than the full version. The gated fusion mechanism provides a small positive contribution in this simpler shared setup, but the deformable residual blocks do not. It shows that the two extra mid-level components should not be treated as a single package whose value can be assumed in advance.

To complement the predictive results, Table 2 reports the computational profile of the same five variants under the shared cross-entropy protocol.

Table 2. Computational profile of the five variants under the shared cross-entropy protocol. A larger version is provided in the Appendix.

Model	Params (M)	Inference Time (s)	Avg Batch Time (s)	Images/s	Peak GPU Mem (MB)
Mid-Level (Full)	73.0513	1.1049	0.1105	18.1011	9610.7344
Mid-Level (No Gating)	73.0513	1.0422	0.1042	19.1904	8621.9727
Mid-Level (No Deform)	72.9683	0.8610	0.0861	23.2290	9612.2075
Late Fusion	69.5363	1.1232	0.1123	17.8056	7943.9741
Early Fusion	36.9279	0.7119	0.0712	28.0935	7104.7778

Early Fusion is the lightest and fastest model, but this efficiency comes with the weakest predictive performance. The dual-branch variants are more computationally demanding, although the extra complexity does not translate into an accuracy advantage for the most elaborate design. In particular, Mid-Level (Full) is among the heaviest models, yet it does not outperform either Late Fusion or Mid-Level (No Deform). This supports one of the central findings of the paper: additional architectural complexity has a cost, but that cost does not automatically lead to a greater robustness.

Overall, the comparison shows that the choice of fusion stage is already highly consequential, that Late Fusion is the most reliable under geographic shift, and that the additional mid-level modules do not produce uniformly positive gains.

5.3. Results Under the Extended Training Recipe

We next evaluate the same five architecture variants under the stronger training recipe described in Section 4. This extended setup adds modality-specific normalisation, synchronised augmentation, Lovasz-based losses, and joint localisation–damage supervision. The corresponding Haiti test results are shown in Table 3.

Table 3. Results under the extended training recipe on the unseen Haiti test set.

Model	Macro F1	mIoU	Pixel Acc.	Kappa
Late Fusion	**0.4312**	**0.3863**	**0.9380**	**0.7240**
Mid-Level (No Gating)	0.4294	0.3834	0.9321	0.7178
Mid-Level (No Deform)	0.4281	0.3817	0.9317	0.7122
Mid-Level (Full)	0.4211	0.3734	0.9348	0.6842
Early Fusion	0.4097	0.3588	0.9246	0.6383

The extended recipe changes the absolute performance levels, but it does not support that the more complex architecture becomes the clear winner. Late Fusion remains the strongest overall model, but the ranking among the three mid-level variants is different. In particular, Mid-Level (No Gating) becomes much more competitive and nearly matches Late Fusion, while the full mid-level model still does not come first, even among all Mid-level models.

Under the simpler shared cross-entropy setup, gating provided a modest positive effect and deformable convolution looks unhelpful. However, under the stronger recipe, the gap between the mid-level variants becomes smaller and the advantage of the more elaborate full configuration still does not emerge. This suggests that optimisation choices may be doing at least as much heavy lifting as the architectural refinements themselves. The training recipe affects performance, but not in a way that consistently favours the more complex mid-level design.

It is also worth noting that the stronger recipe benefits some weaker models more than others. Early Fusion improves substantially relative to its main-protocol result, but it still remains the weakest variant. This again supports the view that training and architecture interact in non-trivial ways: a better optimisation setup can reduce some of the gap, but it does not erase the broader differences between fusion strategies. The corresponding training curves are provided in Appendix Figure A.4.

5.4. Per-Class Performance and Failure Modes

To understand what is driving the aggregate metrics, we next examine class-wise behaviour. Figure 1 shows the per-class IoU under the main cross-entropy protocol, and Figure 2 shows the same comparison under the extended recipe. Larger versions of both figures are included in Appendix Figures A.1 and A.2, while the exact numerical values are reported in Appendix Tables A.4 and A.5.

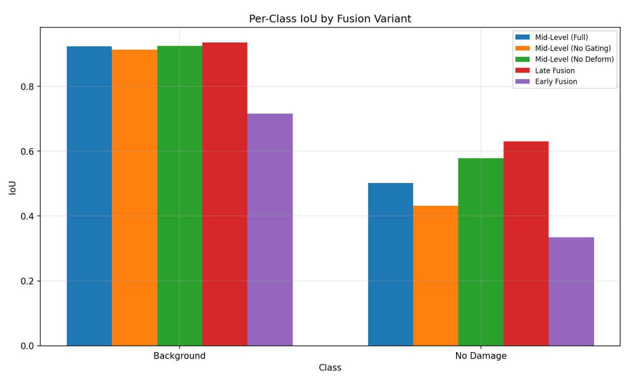

Figure 1. Per-class IoU under the shared cross-entropy protocol. Only classes with non-zero predictions are shown. Minor Damage (0.19%) and Major Damage (0.02%) received zero predictions across all variants. A larger version is provided in the Appendix.

The class-wise analysis reveals a major limitation of all models. In both training setups, the predictions are concentrated almost entirely on the Background and No Damage classes. Minor Damage and Major Damage receive zero predictions across all variants. This means that even when a model achieves a reasonable overall Macro F1 or mIoU, it is still failing to recover the rare but practically important minority damage categories on the Haiti test set.

Among the classes that are actually predicted, Late Fusion performs best overall under the main protocol, especially

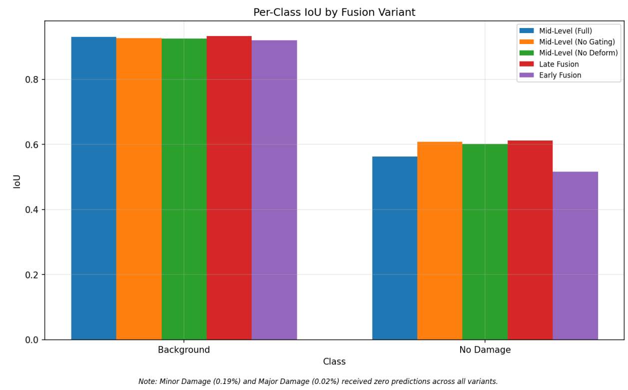

Figure 2. Per-class IoU under the extended training recipe. As in the main comparison, predictions remain concentrated on the Background and No Damage classes. A larger version is provided in the Appendix.

on the No Damage class. Under the extended recipe, the differences between Late Fusion and the stronger mid-level ablations become smaller, but the general picture does not change: the models are still solving the easier part of the problem much better than the harder one. The main unresolved challenge is not simply improving average overlap, but achieving more reliable severity discrimination under class imbalance and domain shift.

5.5. Qualitative Visual Comparison

Figure 3 provides a qualitative example from the Haiti test set under the main shared cross-entropy protocol. A larger version of this example is included in Appendix Figure A.3.

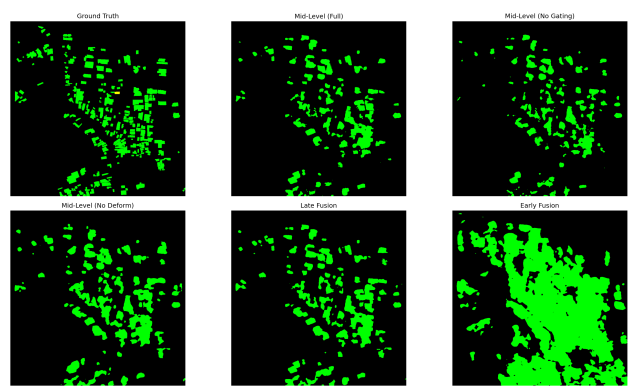

Figure 3. Qualitative comparison on a sample from the Haiti test set under the shared cross-entropy protocol. Predictions are dominated by the No Damage class, with Early Fusion producing noticeably coarser and more merged building regions. A larger version is provided in the Appendix.

The visual comparison is broadly consistent with the quantitative results. Early Fusion produces a much coarser and more merged prediction map, often blending nearby structures into large connected regions. The late- and mid-level variants produce cleaner building footprints and better preserve separation between structures. At the same time, the figure also reinforces the limitation already seen in the per-class analysis: the predictions are overwhelmingly concentrated on the No Damage class, and the finer-grained damage categories are not meaningfully recovered in this example.

The qualitative figure can be mainly used to show the difference in spatial behaviour across fusion strategies. Early Fusion appears much less precise spatially, whereas Late Fusion and the stronger mid-level variants generate more plausible building-level regions while the detailed severity prediction remains weak.

5.6. Summary of Findings

Above all, the results support three main conclusions. First, fusion strategy matters under cross-region generalisation: Early Fusion is consistently the weakest, while Late Fusion is the strongest overall in both the main and extended settings. Second, mid-level fusion remains a meaningful design to study, but its additional components do not show stable gains. Gating helps slightly in the simpler setup, deformable convolution does not, and the stronger training recipe compresses the ranking without making the full mid-level model the clear winner. Third, the most important unresolved difficulty lies at the class level. All variants struggle severely on the minority damage categories, which shows that architectural complexity alone is not enough to deliver robust damage assessment under geographic shift.

6. Conclusion

This project compared multimodal fusion strategies for cross-region building damage assessment from optical and SAR imagery within a shared Transformer-based architecture family. In the dual-branch variants, optical and SAR inputs were processed by architecturally identical but independently parameterised encoders. In addition to the three main fusion settings, the project also examined two targeted ablations of the mid-level design and evaluated the same architecture family under both a simple shared cross-entropy protocol and a stronger training recipe.

The results show that fusion strategy has a clear effect on performance under geographic shift. Across both training setups, Late Fusion remains the strongest overall variant on the unseen Haiti test set, while Early Fusion is consistently the weakest. Mid-level fusion nevertheless remains useful to analyse, because it provides a controlled setting for examining whether added fusion modules actually help. In that comparison, gating offers only a modest benefit in the simpler setup, deformable convolution does not, and the stronger training recipe changes absolute performance without making the full mid-level model the clear winner.

More broadly, results of this project shows that additional architectural complexity does not automatically produce more robust damage assessment. The systematic comparison across fusion settings and ablations is therefore valuable not because it confirms a more complex model is best, but because it clarifies which gains remain stable and which are sensitive to training choices. At the same time, the class-wise analysis makes clear that cross-region damage assessment remains challenging, especially for the rarer damage categories. Future work could build on this comparison by using stronger imbalance-aware objectives, evaluating across more unseen target regions, and placing greater emphasis on fine-grained severity discrimination.

7. Limitations

Several limitations of this project should be acknowledged. First, although the cross-region split used here is stricter and more realistic than a random split, the final evaluation still relies on a single unseen target region, Haiti. In practice, this was a deliberate compromise between evaluation quality and the time and computational cost of retraining and testing multiple multimodal models under different region splits. As a result, the current experiments provide a strong case study of geographic transfer, but they do not yet show how stable the same ranking would be across a wider range of unseen disaster regions.

Second, the main comparison protocol was intentionally kept simple so that the effect of fusion design could be isolated as clearly as possible. Under this setup, training uses cross-entropy loss only on the damage map, while the localisation branch is forward-propagated without its own supervised objective. This simplification is also due to practical constraints: introducing joint supervision, additional loss balancing, and more extensive hyperparameter tuning for all five architecture variants would have required substantially more experimentation time. The extended recipe partly addresses this by adding joint localisation–damage supervision, but the simpler protocol remains the main basis of comparison in the paper.

Third, the experiments make clear that class imbalance and cross-region shift remain difficult unresolved problems. In both training setups, all variants fail to recover the Minor Damage and Major Damage classes on the Haiti test set. This is an important limitation as it shows that reasonable aggregate scores can still hide weak performance on the rarer but practically important damage categories. In other words, the models are currently much better at separating the easier majority classes than at providing reliable fine-grained severity assessment.

Finally, while we use a fixed random seed to keep the experiments reproducible, we do not yet report repeated runs across multiple seeds. This is also because of the cost of training and evaluating several multimodal variants under both the main and extended recipes. For that reason, the relative ordering of closely performing models, especially among the mid-level variants, should be interpreted with some caution. A more ideal and complete follow-up study would include repeated runs, additional held-out regions, and more explicitly imbalance-aware training objectives.

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Appendices

This appendix provides additional material supporting the numerical results in Section 5. It includes class-wise numerical breakdowns under both training setups, enlarged versions of the tables and figures shown in the main text, and the training curves from the extended recipe experiments.

.1. Larger Version of Main-Text Tables

The following tables provide enlarged versions of the main test-set comparisons and the computational efficiency profile reported in Section 5.

Table A.1. Main comparison under the shared cross-entropy protocol on the unseen Haiti test set.

Model	Macro F1	mIoU	Pixel Acc.	Kappa
Late Fusion	**0.4349**	**0.3914**	**0.9403**	**0.7396**
Mid-Level (No Deform)	0.4235	0.3759	0.9301	0.6941
Mid-Level (Full)	0.4072	0.3564	0.9277	0.6296
Mid-Level (No Gating)	0.3895	0.3363	0.9178	0.5602
Early Fusion	0.3339	0.2626	0.7498	0.3847

Table A.2. Computational profile of the five variants under the shared cross-entropy protocol.

Model	Params (M)	Inference Time (s)	Avg Batch Time (s)	Images/s	Peak GPU Mem (MB)
Mid-Level (Full)	73.0513	1.1049	0.1105	18.1011	9610.7344
Mid-Level (No Gating)	73.0513	1.0422	0.1042	19.1904	8621.9727
Mid-Level (No Deform)	72.9683	0.8610	0.0861	23.2290	9612.2075
Late Fusion	69.5363	1.1232	0.1123	17.8056	7943.9741
Early Fusion	36.9279	0.7119	0.0712	28.0935	7104.7778

Table A.3. Results under the extended training recipe on the unseen Haiti test set.

Model	Macro F1	mIoU	Pixel Acc.	Kappa
Late Fusion	**0.4312**	**0.3863**	**0.9380**	**0.7240**
Mid-Level (No Gating)	0.4294	0.3834	0.9321	0.7178
Mid-Level (No Deform)	0.4281	0.3817	0.9317	0.7122
Mid-Level (Full)	0.4211	0.3734	0.9348	0.6842
Early Fusion	0.4097	0.3588	0.9246	0.6383

.2. Per-Class Numerical Breakdown

Tables A.4 and A.5 provide the per-class IoU and F1 values corresponding to the figures shown in the main text. In both settings, the same limitation remains clear: predictions are concentrated on the Background and No Damage classes, while Minor Damage and Major Damage receive zero predictions across all variants.

As in the main text, Minor Damage and Major Damage received zero predictions across all variants in both settings.

.3. Larger Versions of Main-Text Figures

The following figures provide enlarged versions of the class-wise and qualitative comparisons shown in Section 5. These versions make it easier to inspect the relative differences between fusion variants.

Table A.4. Per-class IoU and F1 under the shared cross-entropy protocol.

Model	IoU_Background	IoU_No Damage	F1_Background	F1_No Damage
Mid-Level (Full)	0.9233	0.5024	0.9601	0.6688
Mid-Level (No Gating)	0.9135	0.4317	0.9548	0.6030
Mid-Level (No Deform)	0.9245	0.5790	0.9607	0.7334
Late Fusion	0.9354	0.6302	0.9666	0.7731
Early Fusion	0.7158	0.3345	0.8344	0.5013

Table A.5. Per-class IoU and F1 under the extended training recipe.

Model	IoU_Background	IoU_No Damage	F1_Background	F1_No Damage
Mid-Level (Full)	0.9303	0.5632	0.9639	0.7206
Mid-Level (No Gating)	0.9260	0.6077	0.9616	0.7560
Mid-Level (No Deform)	0.9256	0.6013	0.9614	0.7510
Late Fusion	0.9329	0.6123	0.9653	0.7596
Early Fusion	0.9194	0.5159	0.9580	0.6806

.4. Training Curves Under the Extended Recipe

Figure A.4 shows the training loss, validation loss, and validation mIoU curves for all five architecture variants under the extended training recipe. These curves support the interpretation in Section 5 that the stronger optimisation setup changes absolute performance levels but does not produce a simple reranking in favour of the most complex architecture.

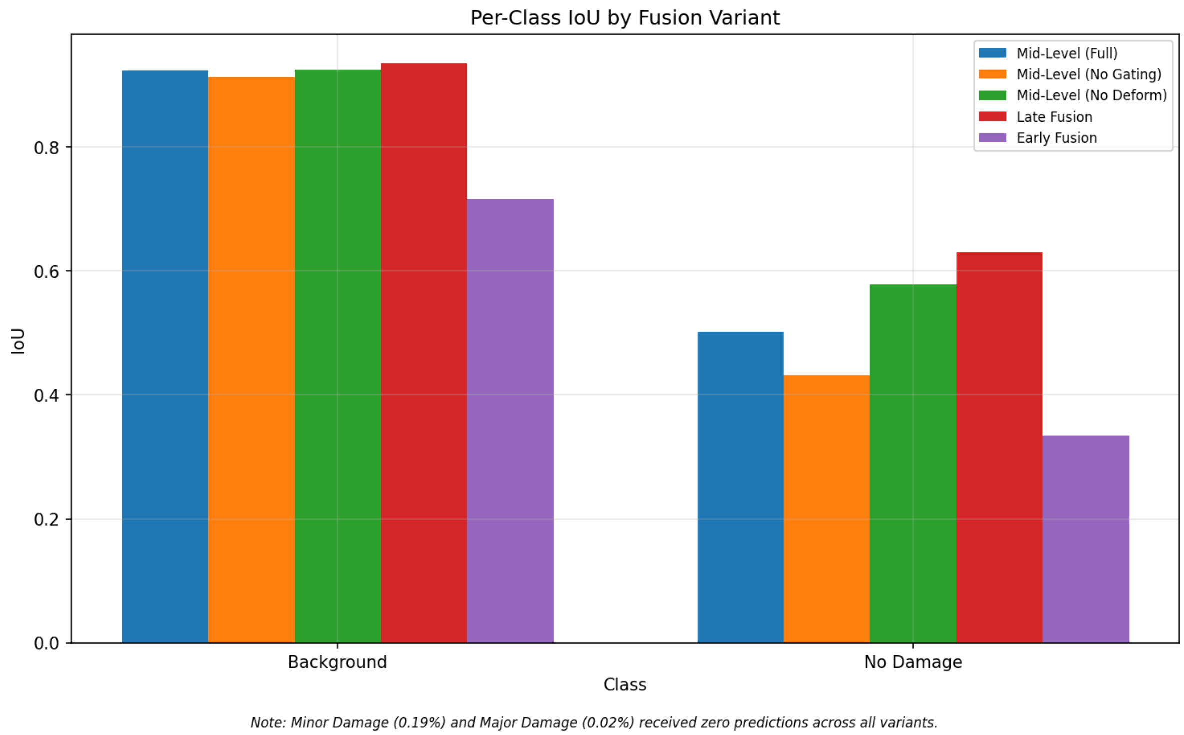

Figure A.1. Larger version of the per-class IoU comparison under the shared cross-entropy protocol. Only classes with non-zero predictions are shown. Minor Damage and Major Damage received zero predictions across all variants.

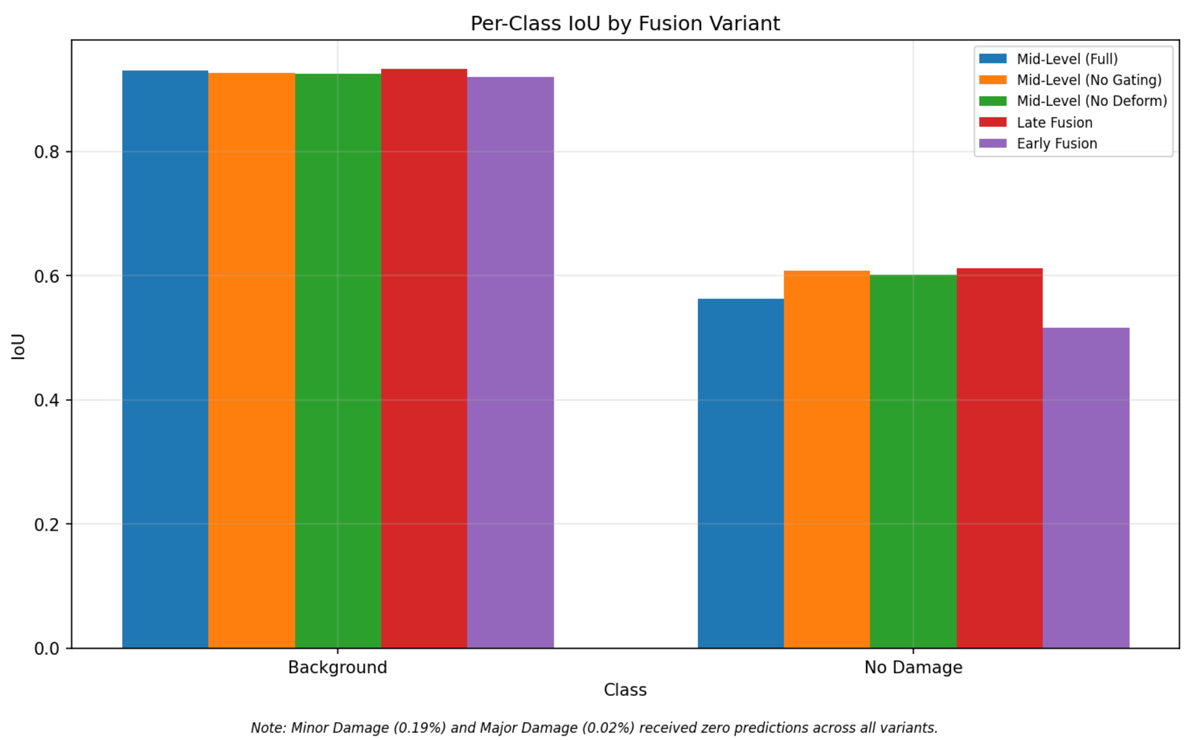

Figure A.2. Larger version of the per-class IoU comparison under the extended training recipe. Predictions remain concentrated on the Background and No Damage classes.

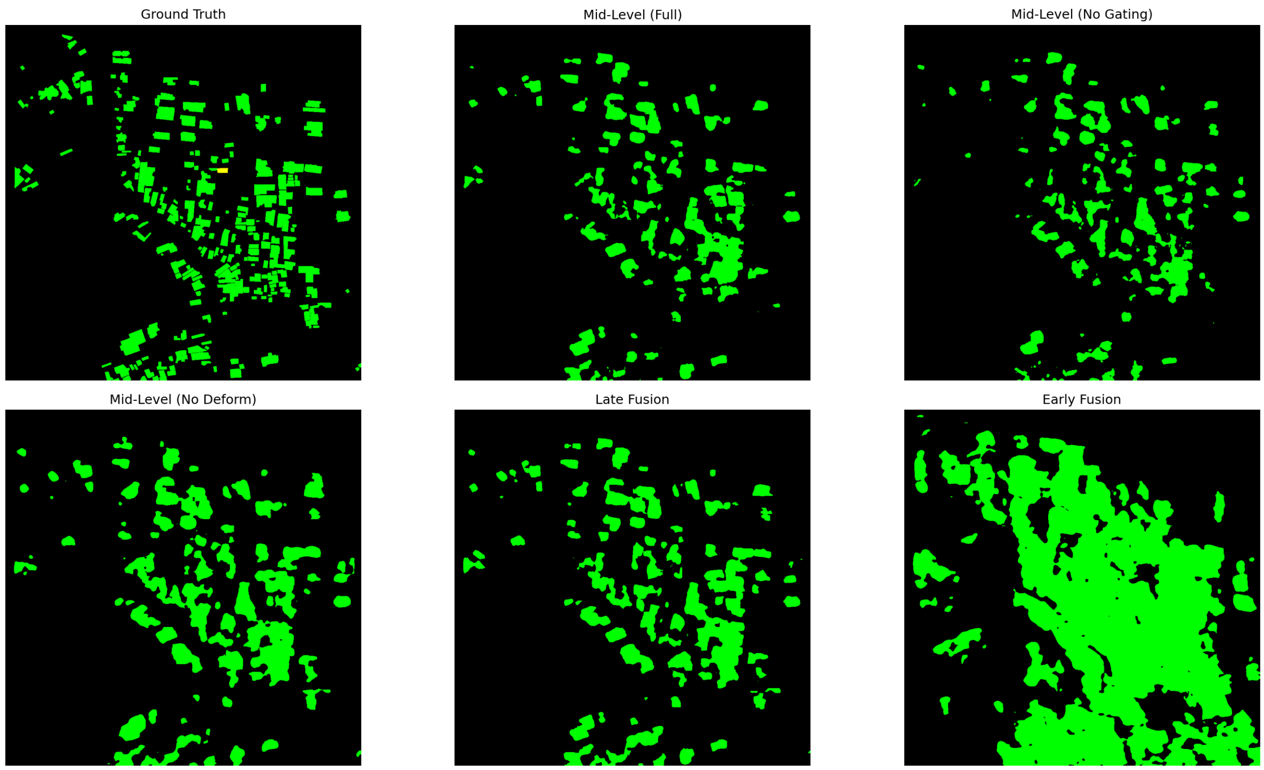

Figure A.3. Larger qualitative comparison on a sample from the Haiti test set under the shared cross-entropy protocol. The enlarged view makes it easier to see the noticeably coarser and more merged predictions produced by Early Fusion, compared with the cleaner spatial behaviour of the stronger late- and mid-level variants.

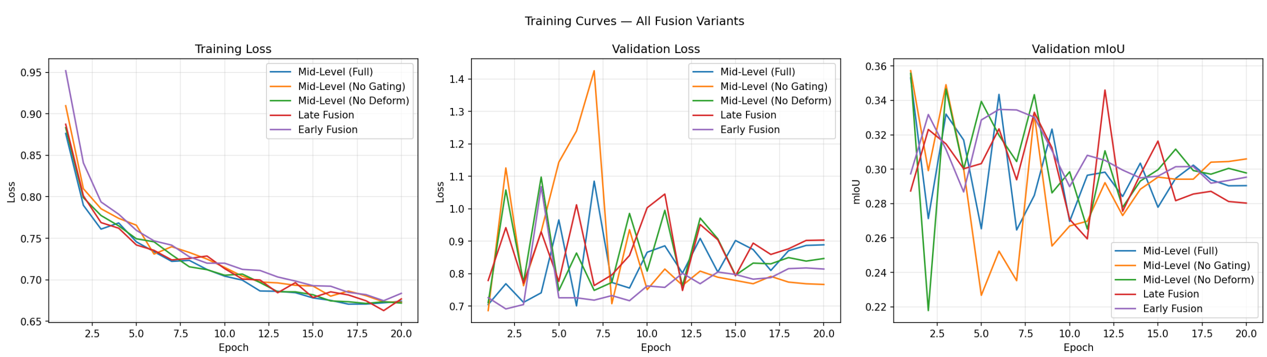

Figure A.4. Training loss, validation loss, and validation mIoU for all five architecture variants under the extended training recipe.